

Team 1

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Engineering Service Learning

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Element A:

Problem Statement: An engineering teacher decides to take a weekend trip with his family, but one of their children has difficulty using the restroom when other people are watching. Since the child is very young and has trouble activating bathroom locks, it's difficult for him to go to the bathroom at other places, especially on vacation. Because using the restroom is an essential human function¹, this problem will occur around thirty times during their trip in various places such as three different airports and two destinations (Davis 2025). This problem is very critical to the well-being of the family during the vacation, as the inconveniences that the child experiences directly influences the happiness that the family feels. Without a solution, the child could experience problems such as lightheadedness², bowel perforation³, or bladder stones⁴. A solution to the inability to operate a bathroom lock is to design a lock that is compatible with numerous different types of doors, complies with TSA carry on regulations⁵, and is operable from the outside of the stall so that the child does not need to interact with the lock to make it hold the door shut.

Problem Elaboration:

An effective bathroom lock would be compatible with all different types of doors, is durable for over thirty uses, is able to withstand around five pounds of force, is less than three pounds, is around the size of the engineering teacher's hand size, and is attachable and detachable in less than five seconds with one hand.

A complete breakdown of the stakeholders involved in our bathroom lock engineering project is provided below:

Our primary stakeholders are the child's father (or the person operating the lock), TSA, and the bathroom stall. The person operating the lock must be able to efficiently and effectively set up the lock with minimal effort from the outside of the stall, and the lock must be able to hold by itself. TSA workers directly have to deal with it to check that it passes airplane guidelines. For this reason, it must be small and not contain hazardous materials while being able to be used in an airport bathroom. The bathroom stall must not be damaged throughout this process, so it must be usable in multiple circumstances.

Secondary stakeholders would be the child, other bathroom occupants, and bathroom maintenance workers. The child will receive the immediate from this prototype and it has to be assured that they don't have to do a thing. This assures the lock has to be usable with multiple doors because of the changing locations. The other occupants are going to see the lock, and if they can't tell what the lock is for, it could cause the lock to be removed by accident and make the child uncomfortable. This means that the lock has to be easily identifiable. They would also have to deal with the complications if the lock does not work properly. There might be inconveniences such as whining that could cause others to be uncomfortable. The bathroom maintenance workers shouldn't be affected, but if the lock gets stuck, creates a mess, or damages the door it could cause problems for them or the family.

Our tertiary stakeholders could be other families with family members with similar issues and the child's family (not using the lock). If another family has someone with similar issues, this could cause them to get or create a similar device. The child's family also is affected because this allows them to go to a variety of places. This means that they get to have flexibility with plans

and the ability to go on vacations. The engineering classroom is also affected because a large amount of the materials used to build the lock will come from materials inside of the engineering classroom. The owners of the bathroom facility could also be affected, because if there are damages to the bathroom stall, the owner has to organize and pay for the repairs

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